CS-GY 6923: Lecture 8
Kernel Methods

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Non-linear methods

- Previous methods studied (regression, logistic regression) are considered linear methods.
- They make predictions based on $\langle \mathbf{x}, \boldsymbol{\beta} \rangle$ – i.e. based on weighted sums of features.
- Next part of the course: we move on to non-linear methods. Specifically, **kernel methods** and **neural networks**.
- Both are very closely related to feature transformations, which was one technique we saw for using linear methods to learn non-linear concepts.

Recall: k-nearest neighbor method

k-NN algorithm: a simple but powerful baseline for classification.

Training data: $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$ where $y_1, \ldots, y_n \in \{1, \ldots, q\}$.

Classification algorithm:

Given new input $\mathbf{x}_{new}$,

- Compute $sim(\mathbf{x}_{new}, \mathbf{x}_1), \ldots, sim(\mathbf{x}_{new}, \mathbf{x}_n)$.[1]
- Let $\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}$ be the training data vectors with highest similarity to $\mathbf{x}_{new}$.
- Predict y_{new} as $majority(y_{j_1}, \ldots, y_{j_k})$.

[1] $sim(\mathbf{x}_{new}, \mathbf{x}_i)$ is any chosen similarity function, like $1 - \|\mathbf{x}_{new} - \mathbf{x}_i\|_2$.

k-nearest neighbor method

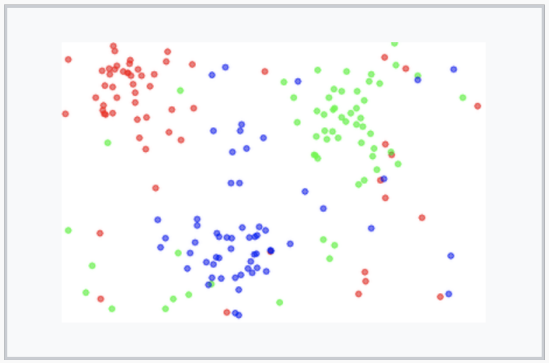


Fig. 1. The dataset.

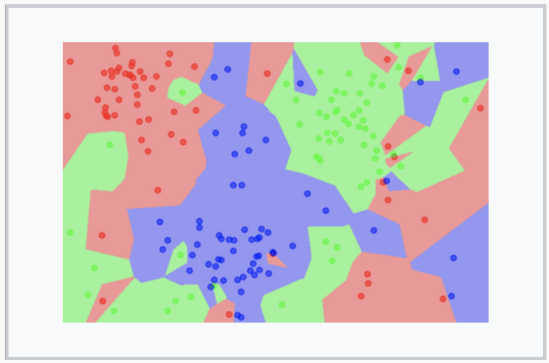


Fig. 2. The 1NN classification map.

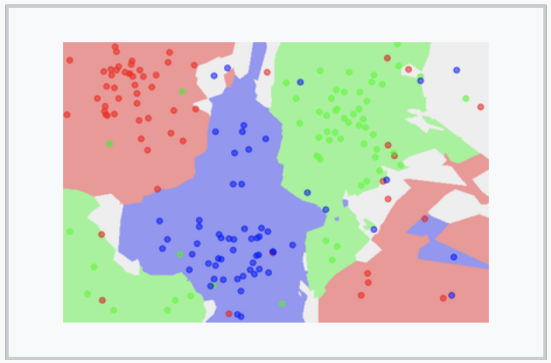


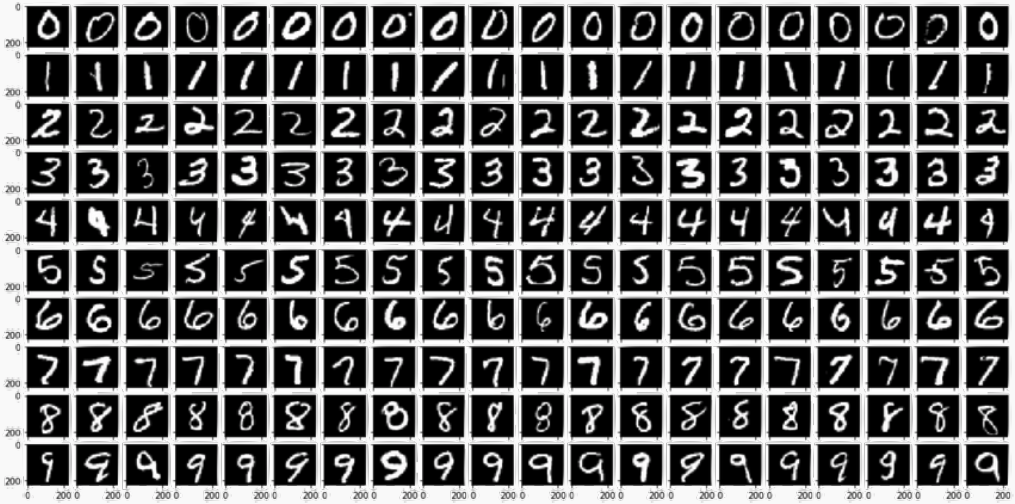
Fig. 3. The 5NN classification map.

- Smaller k, more complex classification function.
- Larger k, more robust to noisy labels.

Works remarkably well for many datasets.

MNIST image data

Especially good for large datasets with lots of repetition. Works well on MNIST for example:



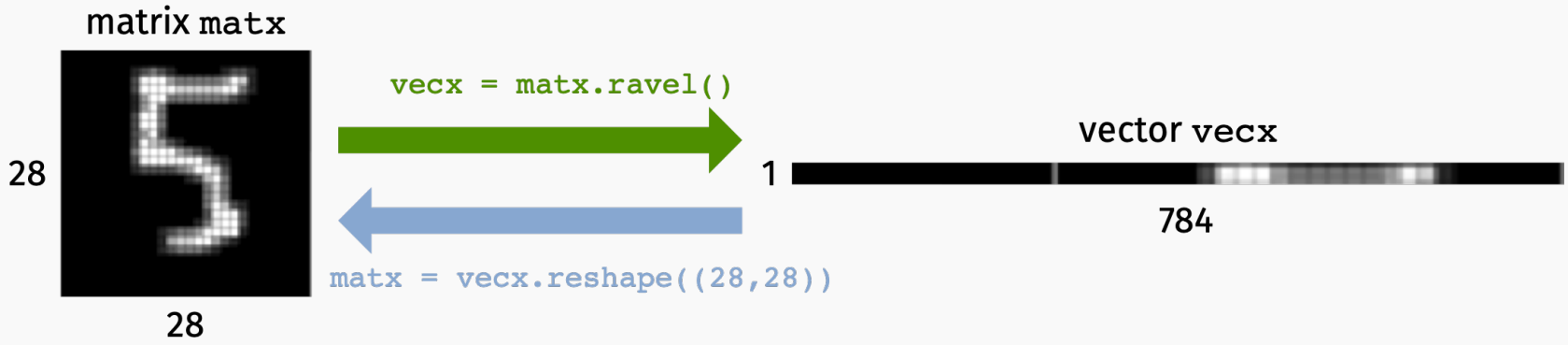
$\approx 95\%$ **Accuracy out-of-the-box.**[2]

Let's look into this example a bit more...

[2] Can be improved to 99.5% with a fancy similarity function!

MNIST image data

Each pixel is number from $[0, 1]$. 0 is black, 1 is white. Represent 28×28 matrix of pixel values as a flattened vector.




```
xmat = np.array([[1,2,3],[4,5,6],[7,8,9]])
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

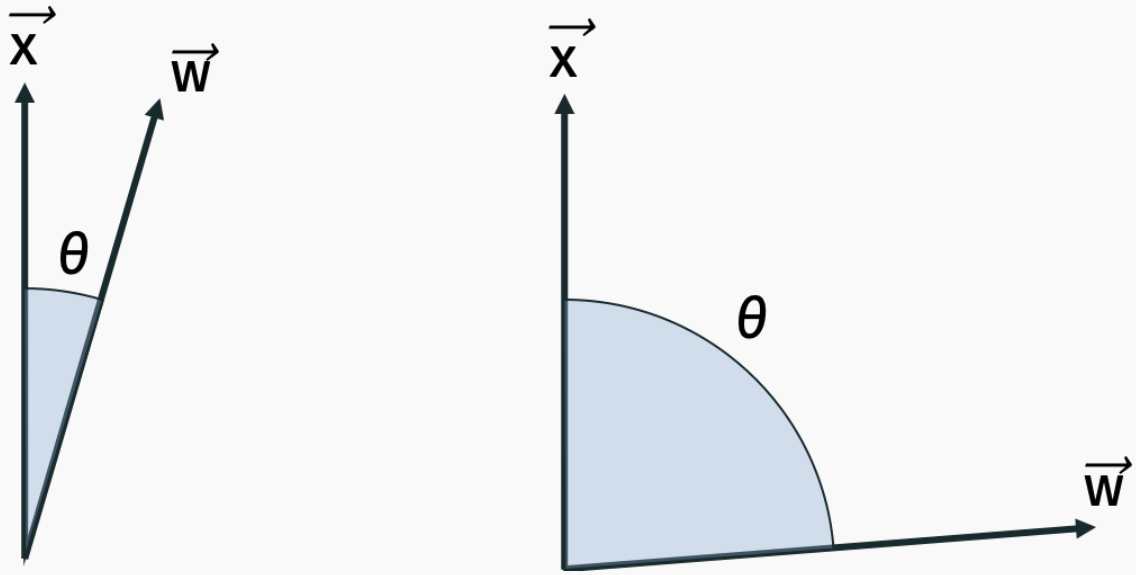
```
xvec = xmat.ravel()
```

```
array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Inner product similarity

Given data vectors $\mathbf{x}, \mathbf{w} \in \mathbb{R}^d$, the inner product $\langle \mathbf{x}, \mathbf{w} \rangle$ is a natural similarity measure.

$$\langle \mathbf{x}, \mathbf{w} \rangle = \sum_{i=1}^{d} x_i w_i = \cos(\theta) \|\mathbf{x}\|_2 \|\mathbf{w}\|_2.$$



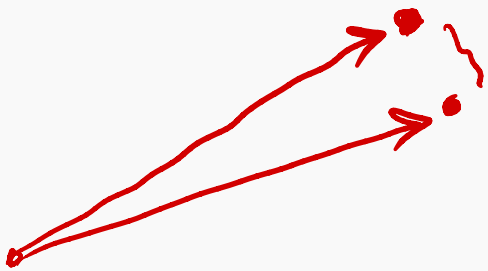
Also called "cosine similarity".

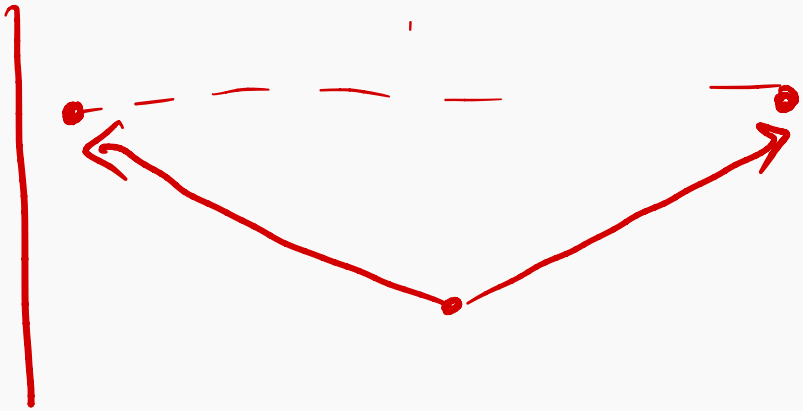
Inner product similarity

Connection to Euclidean (ℓ_2) Distance:

$$\|\mathbf{x} - \mathbf{w}\|_2^2 = \|\mathbf{x}\|_2^2 + \|\mathbf{w}\|_2^2 - 2\langle \mathbf{x}, \mathbf{w} \rangle$$

If all data vectors has the same norm, the pair of vectors with largest inner product is the pair with smallest Euclidean distance.





Inner product for mnist

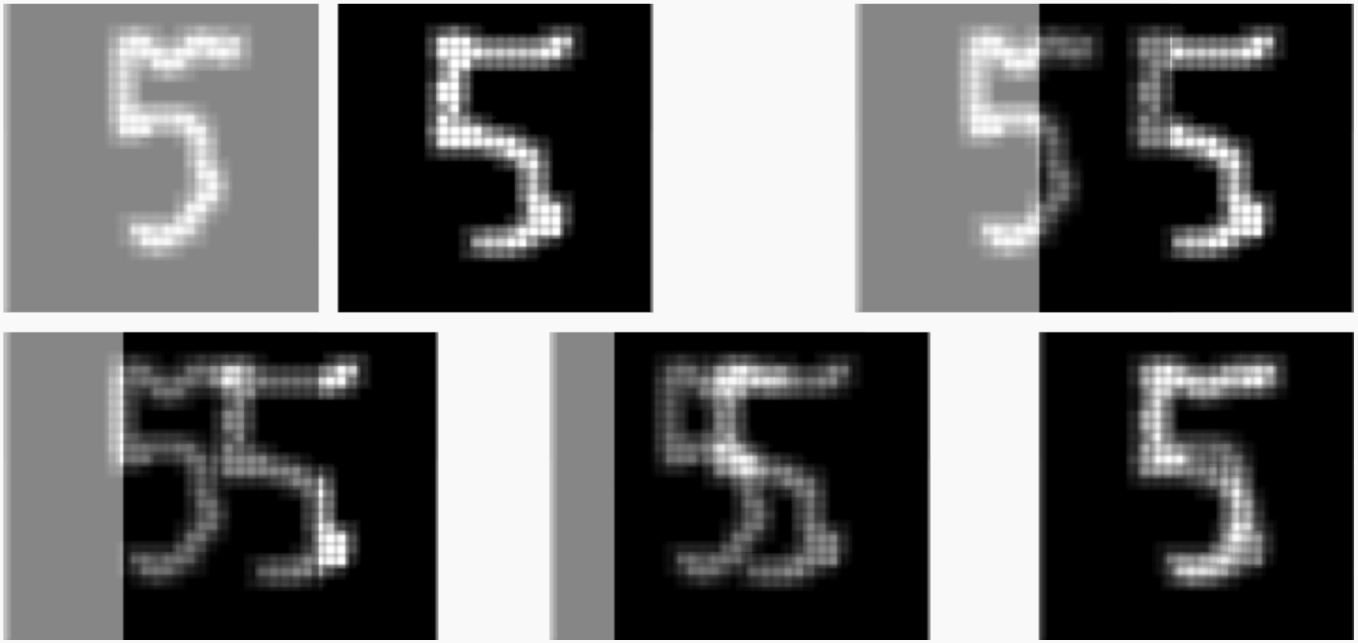
Inner product between MNIST digits:

$$\langle \mathbf{x}, \mathbf{w} \rangle = \sum_{i=1}^{28} \sum_{j=1}^{28} \texttt{matx}[i,j] \cdot \texttt{matw}[i,j].$$

Inner product similarity is higher when the images have large pixel values (close to 1) in the same locations. I.e. when they have a lot of overlapping white/light gray pixels.

Inner product for mnist

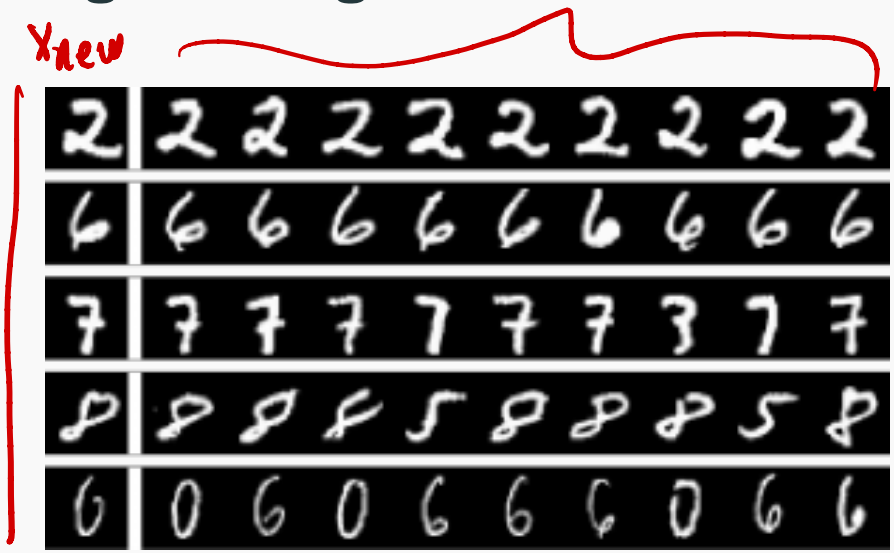
Visualizing the inner product between two images:



Images with high inner product have a lot of overlap.

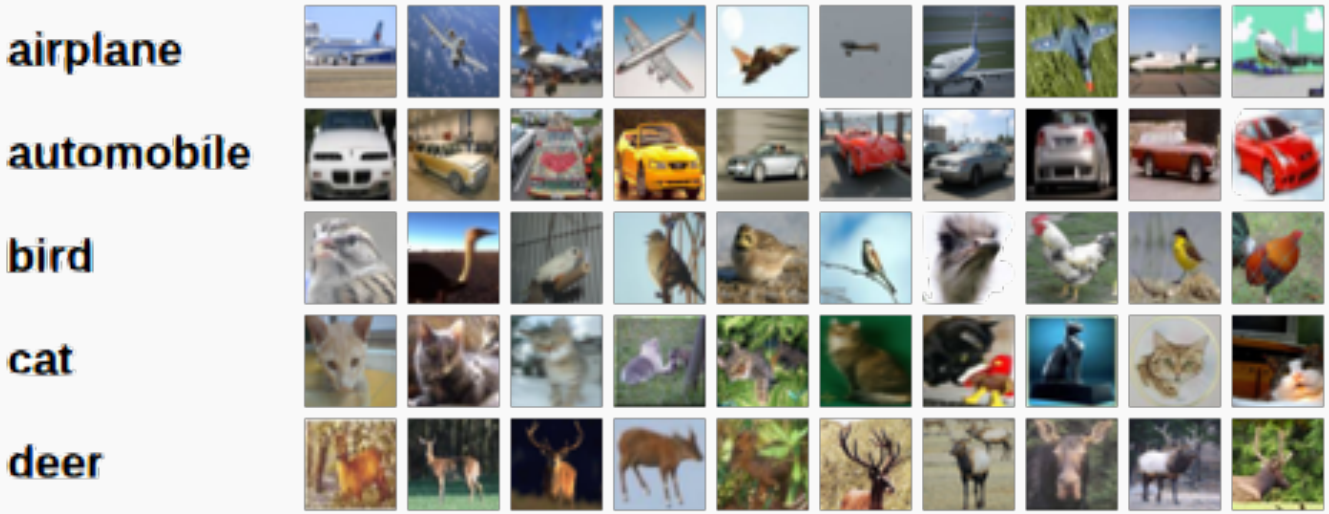
k-NN algorithm on MNIST

Most similar images during k-NN search, $k = 9$:



k-NN for other images

Does not work as well for less standardized classes of images:



CIFAR 10 Images

Even after scaling to have same size, converting to separate RGB channels, etc. something as simple as k-NN won't work.

Another view on logistic regression

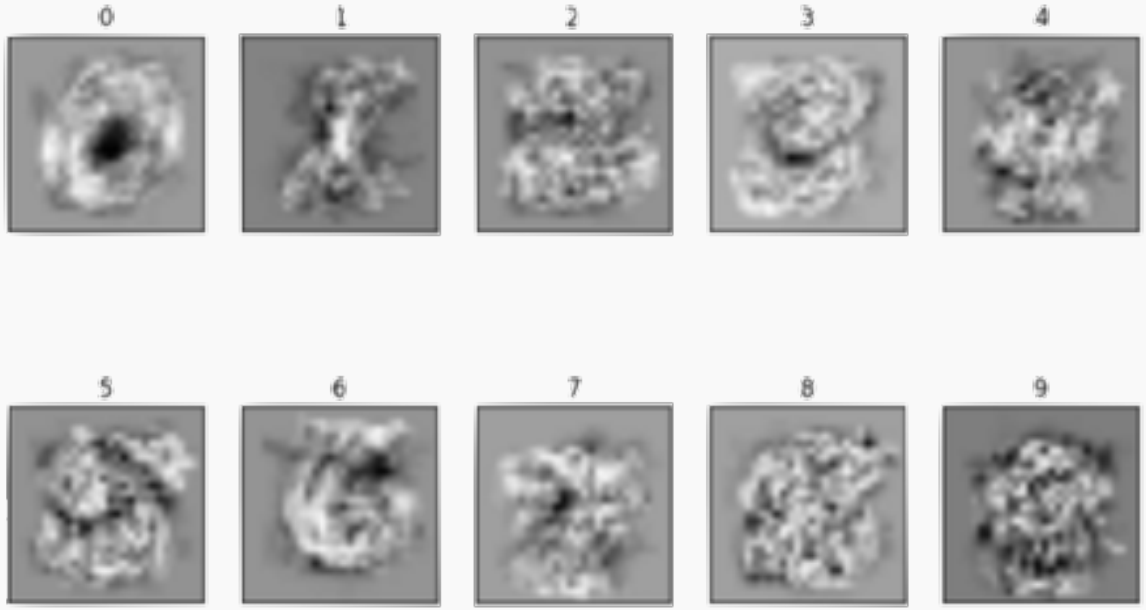
One-vs.-all or Multiclass Cross-entropy Classification with Logistic Regression:

- Learn q classifiers with parameters $\boldsymbol{\beta}^{(1)}, \boldsymbol{\beta}^{(2)}, \ldots, \boldsymbol{\beta}^{(q)}$.
- Given $\mathbf{x}_{new}$ compute $\langle \mathbf{x}_{new}, \boldsymbol{\beta}^{(1)} \rangle, \ldots, \langle \mathbf{x}_{new}, \boldsymbol{\beta}^{(q)} \rangle$
- Predict class $y_{new} = \arg\max_i \langle \mathbf{x}_{new}, \boldsymbol{\beta}^{(i)} \rangle$.

If each $\mathbf{x}$ is a vector with $28 \times 28 = 784$ entries than each $\boldsymbol{\beta}^{(i)}$ also has 784 entries. Each parameter vector can be viewed as a 28×28 image.

Matched filter

Visualizing $\beta^{(0)}, \ldots, \beta^{(9)}$:

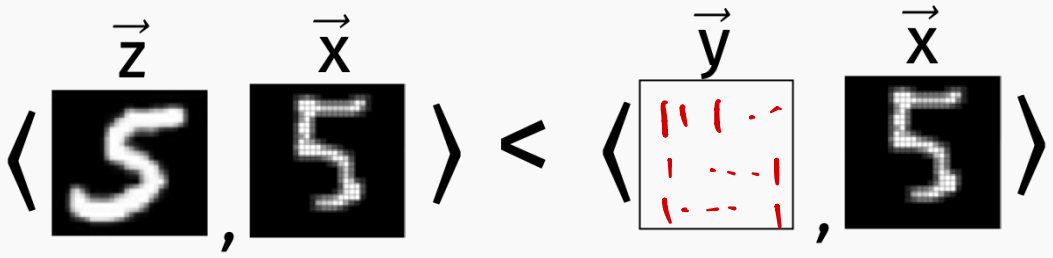


Logistic regression classification rule: For an input  , compute inner product similarity with all weight matrices and choose most similar one.

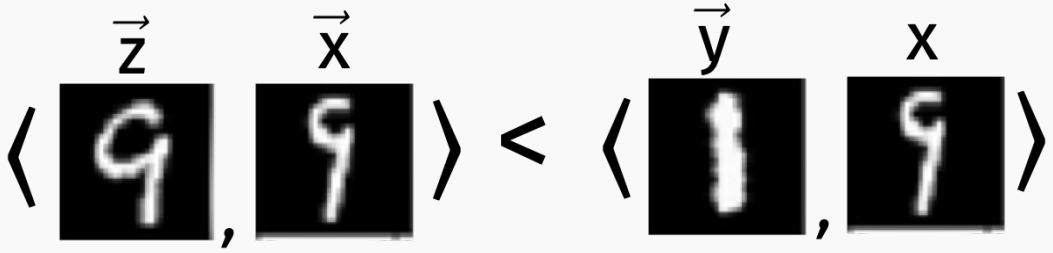
In contrast to k-NN, only need to compute similarity with 10 items instead of n.

Diving into similarity

Often the inner product does not make sense as a similarity measure between data vectors. Here's an example (recall that smaller inner product means less similar):



But clearly the first image is more similar.



Here's a more realistic scenario.

Kernel functions: a new measure of similarity

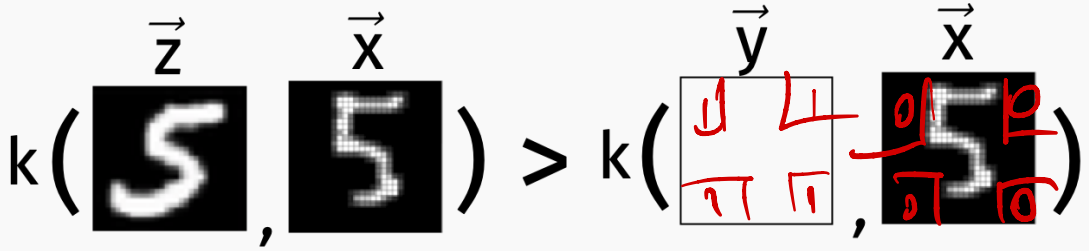
A kernel function $k(\mathbf{x}, \mathbf{y})$ is simply a similarity measure between data points.

$$k(\mathbf{x}, \mathbf{y}) = \begin{cases} \text{large if } \mathbf{x} \text{ and } \mathbf{y} \text{ are similar.} \\ \text{close to 0 if } \mathbf{x} \text{ and } \mathbf{y} \text{ are different.} \end{cases}$$

Example: The Radial Basis Function (RBF) kernel, aka the Gaussian kernel:

$$k(\mathbf{x}, \mathbf{y}) = e^{-\|\mathbf{x}-\mathbf{y}\|_2^2/\sigma^2}$$

for some scaling factor σ.



Kernel functions: a new measure of similarity

Lots of kernel functions involve transformations of $\langle \mathbf{x}, \mathbf{y} \rangle$ or $\|\mathbf{x} - \mathbf{y}\|_2$:

- Gaussian RBF Kernel: $k(\mathbf{x}, \mathbf{y}) = e^{-\|\mathbf{x}-\mathbf{y}\|_2^2/\sigma^2}$
- Laplace Kernel: $k(\mathbf{x}, \mathbf{y}) = e^{-\|\mathbf{x}-\mathbf{y}\|_1/\sigma}$
- Polynomial Kernel: $k(\mathbf{x}, \mathbf{y}) = (\langle \mathbf{x}, \mathbf{y} \rangle + 1)^q$.

But you can imagine much more complex similarity metrics.

How to use a kernel function?

For k-nearest neighbors, can easily replace inner product with whatever similarity function you want.

For logistic regression, it is less clear how to do so.

How to use a kernel function?

Logistic Regression Loss:

$$L(\beta^{(1)}, \ldots, \beta^{(q)}) = -\sum_{i=1}^{n}\sum_{\ell=1}^{q} \mathbb{1}[y_i = \ell] \cdot \log \frac{e^{\langle \beta^{(\ell)}, \mathbf{x}_i \rangle}}{\sum_{j=1}^{q} e^{\langle \beta^{(j)}, \mathbf{x}_i \rangle}}$$

Loss inherently involves inner product between each $\beta^{(j)}$ and each data vector $\mathbf{x}_i$.

Solution: Only work with similarity metrics that can be expressed as inner products.

Kernel functions from feature transformation

A positive semidefinite (PSD) kernel is any similarity function with the following form:

$$k(\mathbf{x}, \mathbf{w}) = \phi(\mathbf{x})^T \phi(\mathbf{w})$$

where $\phi : \mathbb{R}^d \to \mathbb{R}^m$ is a some feature transformation function.

Kernel functions and feature transformation

Example: Degree 2 polynomial kernel, $k(\mathbf{x}, \mathbf{w}) = (\mathbf{x}^T\mathbf{w} + 1)^2$.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \phi(\mathbf{x}) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ \sqrt{2}x_3 \\ x_1^2 \\ x_2^2 \\ x_3^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}x_1x_3 \\ \sqrt{2}x_2x_3 \end{bmatrix}$$

$\phi: \mathbb{R}^3 \to \mathbb{R}^{10}$

$\phi(w) = \begin{bmatrix} 1 \\ \sqrt{2}w_1 \\ \sqrt{2}w_2 \\ \vdots \end{bmatrix}$

$$\begin{aligned}
(\mathbf{x}^T\mathbf{w} + 1)^2 &= (x_1w_1 + x_2w_2 + x_3w_3 + 1)^2 \\
&= 1 + 2x_1w_1 + 2x_2w_2 + 2x_3w_3 + x_1^2w_1^2 + x_2^2w_2^2 + x_3^2w_3^2 \\
&\quad + 2x_1w_1x_2w_2 + 2x_1w_1x_3w_3 + 2x_2w_2x_3w_3 = \phi(x)^T \cdot \phi(w) \\
&= \phi(\mathbf{x})^T\phi(\mathbf{w}).
\end{aligned}$$

Kernel functions and feature transformation

Not all similarity metrics are positive semidefinite (PSD), but all of the ones we saw earlier are:

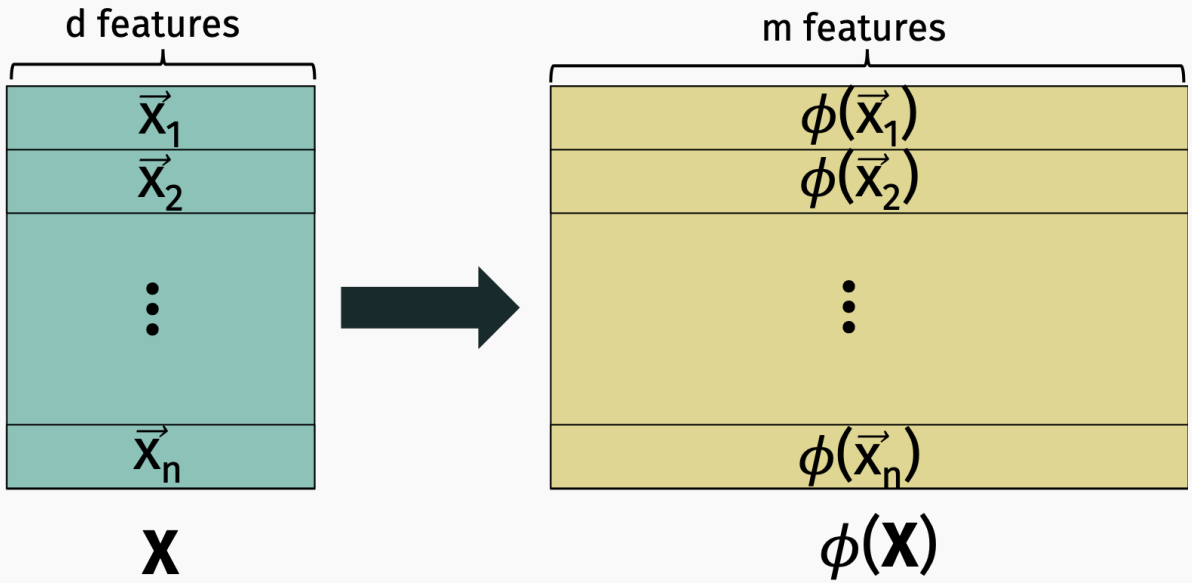
- Gaussian RBF Kernel: $k(\mathbf{x}, \mathbf{y}) = e^{-\|\mathbf{x}-\mathbf{y}\|_2^2/\sigma^2}$
- Laplace Kernel: $k(\mathbf{x}, \mathbf{y}) = e^{-\|\mathbf{x}-\mathbf{y}\|_2/\sigma}$
- Polynomial Kernel: $k(\mathbf{x}, \mathbf{y}) = (\langle \mathbf{x}, \mathbf{y} \rangle + 1)^q$.

And there are many more...

Kernel functions and feature transformation

Feature transformations $\Longleftrightarrow$ new similarity metrics.

To work with the similarity $k(\cdot, \cdot)$ in place of the inner product $\langle \cdot, \cdot \rangle$, it suffices to replace every data point $\mathbf{x}_1, \ldots, \mathbf{x}_n$ by $\phi(\mathbf{x}_1), \ldots, \phi(\mathbf{x}_n)$.



Kernel functions and feature transformation

There are two major issues with this:

- While $\phi(\mathbf{x})$ is sometimes simple and explicit. **More often, it is not.** We might be able to show a kernel is PSD without easily being able to write down $\phi(\mathbf{x})$.
- Transform dimension m is often very large: e.g. $m = O(d^q)$ for a degree q polynomial kernel. For many kernels (e.g. the Gaussian kernel) m is actually *infinite*.

$\phi: \mathbb{R}^d \to \mathbb{R}^m$

So doing the feature transformation explicitly would have very high computational cost. Ideally we would like algorithms that run in better then $O(\infty)$ time.

Reparameterization trick

For simplicity, let's just consider the binary cross entropy/logistic regression loss:

$$L(\beta) = -\sum_{j=1}^{n} y_j \log(h(\mathbf{X}\beta)_j) + (1 - y_j) \log(1 - h(\mathbf{X}\beta)_j)$$

where $h(z) = \frac{1}{1+e^{-z}}$.

Reparameterization trick

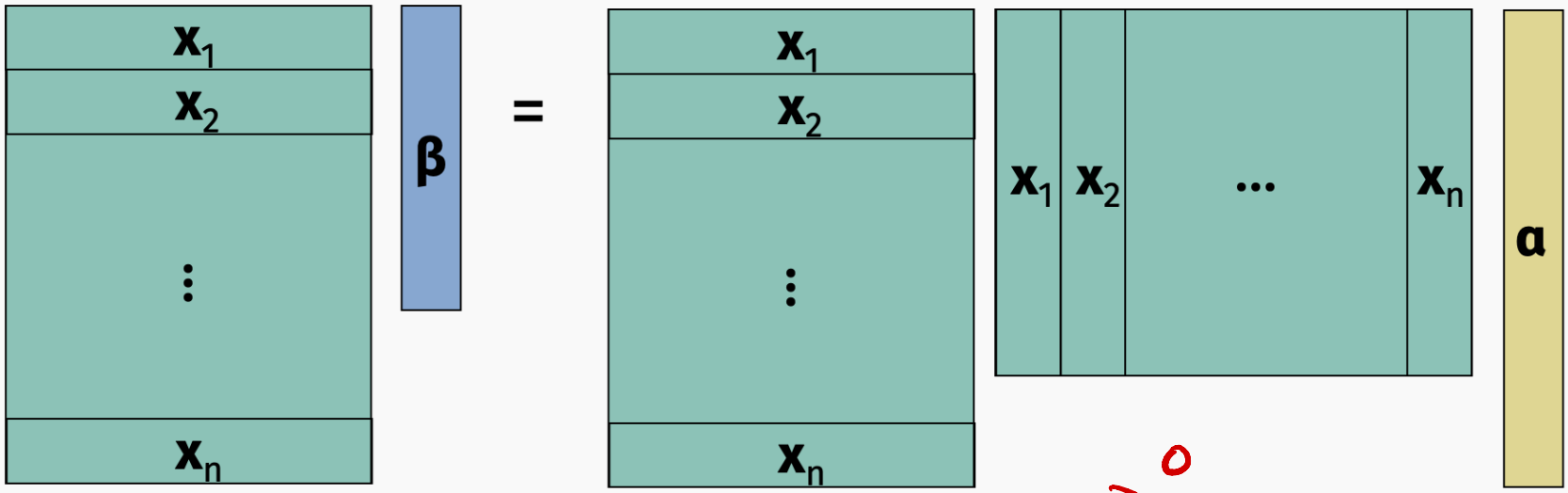
Reminder from linear algebra: Without loss of generality, can assume that β lies in the row span of $\mathbf{X}$.

$\beta = \sum \alpha_i X_i$

So for any $\beta \in \mathbb{R}^d$, there exists a vector $\alpha \in \mathbb{R}^n$ such that:

$\beta = X^T \alpha$

$$\mathbf{X}\beta = \mathbf{X}\mathbf{X}^T\alpha.$$



$X\beta = X(r+a) = Xr + Xa^{0} = Xr$

Reparameterization trick

Logistic Regression Equivalent Formulation: Given data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and binary label vector $\mathbf{y} \in \{0, 1\}^n$ for class i, find $\boldsymbol{\alpha} \in \mathbb{R}^n$ to minimize the loss:

$$L(\alpha) = -\sum_{j=1}^{n} y_j \log(h(\mathbf{X}\mathbf{X}^T\boldsymbol{\alpha})_j) + (1 - y_j) \log(1 - h(\mathbf{X}\mathbf{X}^T\boldsymbol{\alpha})_j)$$

Can still be minimized via gradient descent:

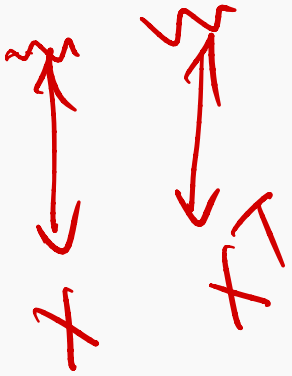
$$\nabla L(\boldsymbol{\alpha}) = \mathbf{X}\mathbf{X}^T(h(\mathbf{X}\mathbf{X}^T\boldsymbol{\alpha}) - \mathbf{y}).$$

$\downarrow X\beta$

Reparameterization trick

If we use a non-linear data transformation ϕ (corresponding to a PSD kernel), then the loss is:

$$-\sum_{j=1}^{n} y_j \log(h(\phi(\mathbf{X})\phi(\mathbf{X})^T\boldsymbol{\alpha})_j) + (1-y_j)\log(1-h(\phi(\mathbf{X})\phi(\mathbf{X})^T\boldsymbol{\alpha})_j)$$

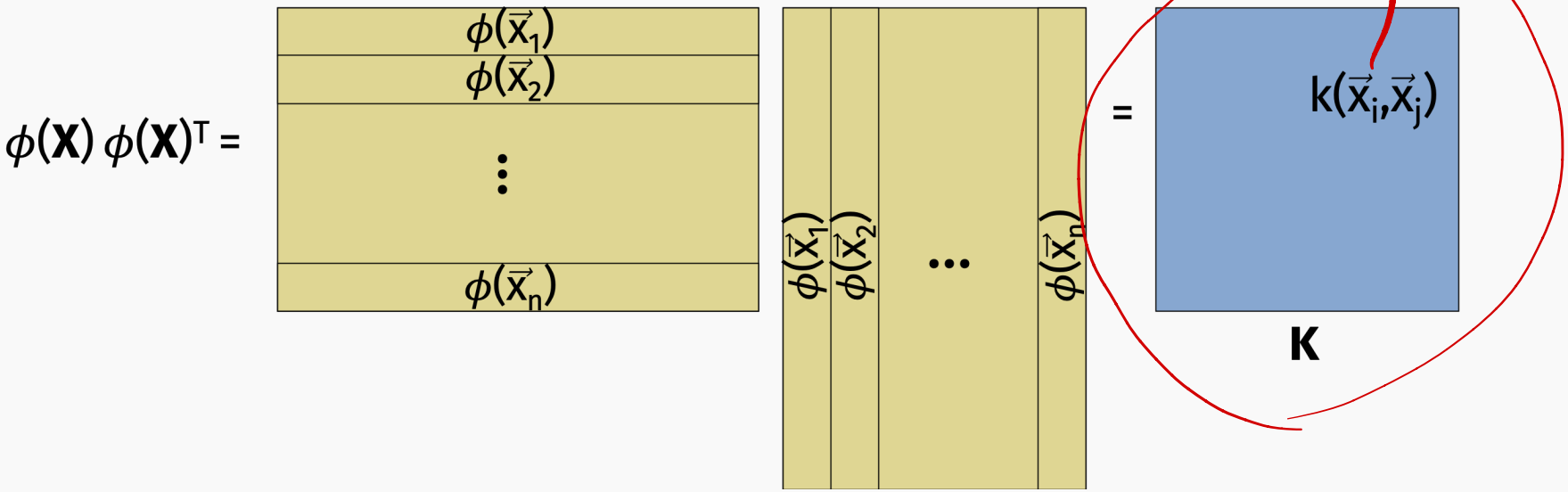


Kernel matrix

$K(x, w) = \phi^T(x)\,\phi(w)$

$= \phi^T(x_i)\,\phi(x_j)$

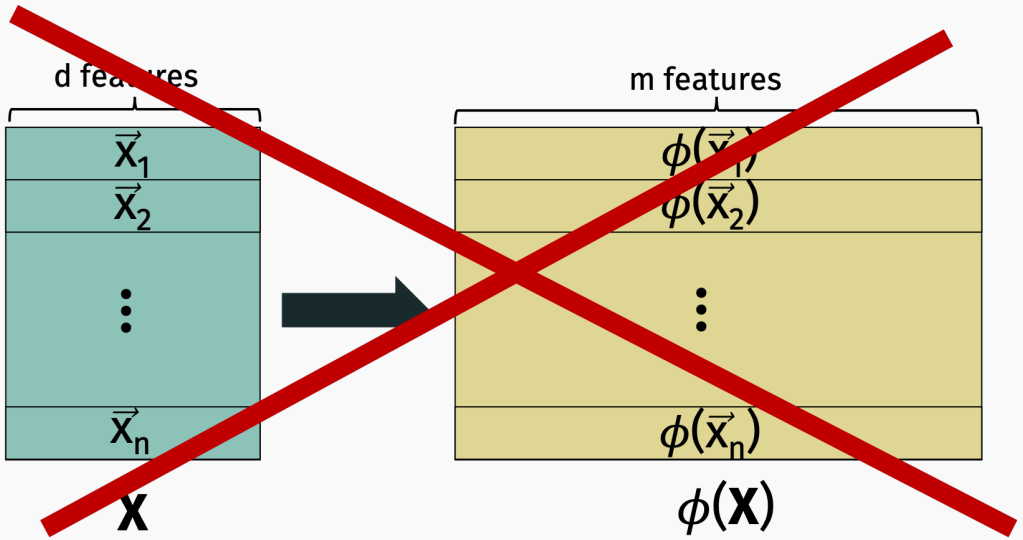
$\mathbf{K} = \phi(\mathbf{X})\phi(\mathbf{X})^T$ is called the kernel Gram matrix.



Kernel trick

We never need to actually compute $\phi(\mathbf{x}_1), \ldots, \phi(\mathbf{x}_n)$ explicitly!

- For training we just need the kernel matrix $\mathbf{K}$, which requires computing $k(\mathbf{x}_i, \mathbf{x}_j)$ for all i, j.



We can always work with a finite sized $n \times n$ matrix.

Kernel trick

Take away:

- Logistic regression can be combined with any positive semidefinite kernel matrix, and the model can be trained in time independent of the transform dimension m.

Kernel trick: prediction

Prediction:

- Prediction can also be done efficiently. For a new input $\mathbf{x}_{new}$, we need to compute:

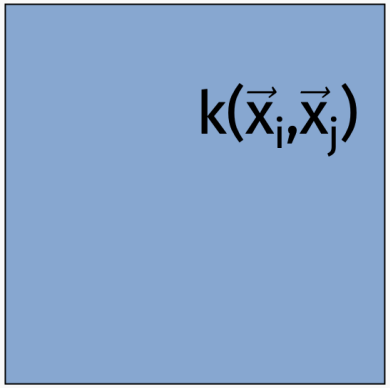
$$\langle \phi(\mathbf{x}_{new}), \boldsymbol{\beta} \rangle = \langle \phi(\mathbf{x}_{new}), \phi(\mathbf{X})^T \boldsymbol{\alpha} \rangle$$
$$= \langle \phi(\mathbf{x}_{new}), \sum_{i=1}^{n} \phi(\mathbf{x}_i)\alpha_i \rangle = \sum_{i=1}^{n} \alpha_i \langle \phi(\mathbf{x}_{new}), \phi(\mathbf{x}_i) \rangle.$$

$k(x_{new}, x_i)$

Each term in the sum $\langle \phi(\mathbf{x}_{new}), \phi(\mathbf{x}_i) \rangle = k(\mathbf{x}_{new}, \mathbf{x}_i)$ can be computed without explicit feature transformation.

Beyond the kernel trick

The kernel matrix $\mathbf{K}$ is still $n \times n$ though which is huge when the size of the training set n is large. Has made the kernel trick less appealing in some modern ML applications.

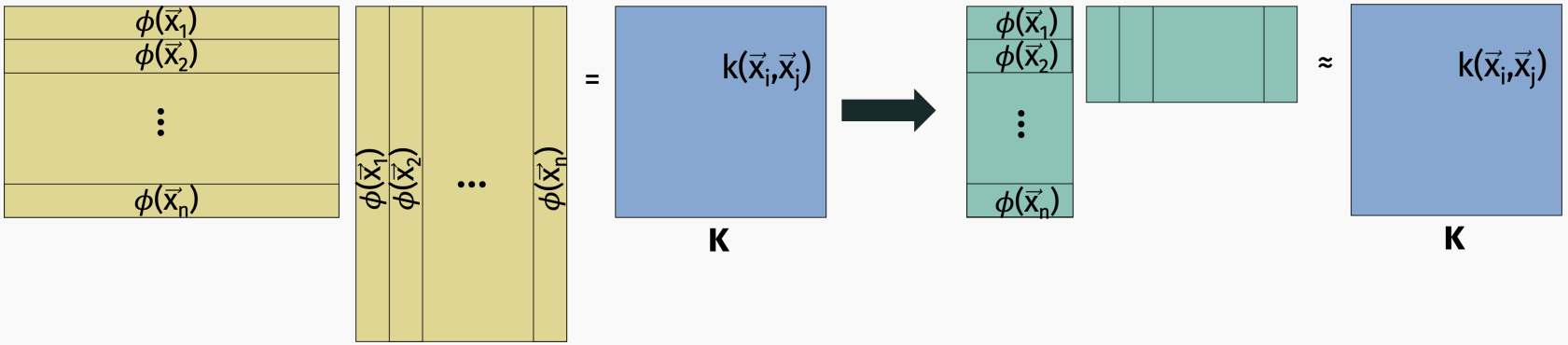


There is an inherent quadratic dependence on n in the computational and space complexity of kernel methods.

- $10,000$ data points $\rightarrow$ runtime scales as $\sim 100,000,000$, $\mathbf{K}$ takes 800MB of space.
- $1,000,000$ data points $\rightarrow$ runtime scales as $\sim 10^{12}$, $\mathbf{K}$ takes 8TB of space.

Beyond the kernel trick

Many algorithmic advances in recent years partially address this computational challenge (random Fourier features methods, Nystrom methods, etc.)



Kernel regression

The kernel trick can also be applied outside of classification. E.g. to regression:

$$\min_{\boldsymbol{\beta}} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2 \rightarrow \min_{\boldsymbol{\alpha}} \|\mathbf{X}\mathbf{X}^T\boldsymbol{\alpha} - \mathbf{y}\|_2^2 + \lambda\|\mathbf{X}^T\boldsymbol{\alpha}\|_2^2$$

Replace $\mathbf{X}\mathbf{X}^T$ by kernel matrix $\mathbf{K}$ during training.

Prediction:

$$y_{new} = \sum_{i=1}^{n} \alpha_i \cdot k(\mathbf{x}_{new}, \mathbf{x}_i).$$

Added benefit: Relatively numerically stable. E.g. is a much better option for performing multivariate or even single variate polynomial regression than direct feature expansion.